```
2  LOGIC_GND     Logická zem
3  NC   Rezerva
4  NC   Rezerva
5  24V_BATT      Batériová vetva 24 V
6  5V_BATT       Batériová vetva 5 V
7  BATT_GND      Batériová zem
8  BATT_GND      Batériová zem
9  5V_LOGIC      LOGIC 5 V
0  LOGIC_GND     Logická zem
1  NC   Rezerva
2  NC   Rezerva
3  I2C_SCL   I²C hodinový signál
4  I2C_SDA   I²C dátový signál
5  NC   Rezerva pre budúce IRQ
6  NC   Rezerva
7  SPI_SCK   SPI clock
8  LOGIC_GND     Referenčná zem SPI
9  SPI_MOSI      SPI controller → periféria
0  SPI_MISO      SPI periféria → controller
1  SPI_CS0   Výber SPI zariadenia 0
2  SPI_CS1   Výber SPI zariadenia 1
3  SPI_DC    Display Data/Command
4  LOGIC_GND     Referenčná zem SPI
5  NC   Rezerva
6  RS485_GND     Referenčná zem RS-485
7  RS485_A   Diferenciálna linka A
8  RS485_B   Diferenciálna linka B
9  NC   Rezerva
0  NC   Rezerva
```

BACKBONE V2.1
Medulla Oblongata

H101
M20-6102045

24V, GND, 5V
LOGIC_GND
24V_BATT, BATT_GND, 5V_LOGIC
I2C_SCL, I2C_SDA
SPI_SCK, SPI_MOSI, SPI_DC, SPI_SC1, UART_TX, RS485_A, UART_RX
5V_BATT, SPI_MISO, SPI_CS2, RS485_GND, RS485_B

R120
MBB02070Z0000ZC100
24V_LOGIC

MANUS

IN LINE PULL UP ULNA

R102 ERJPA3F4701V — LOGIC_3V3
R103 ERJ-3EKF1001V — INDEX
C102 C0603C103K5RACTU — LOGIC_GND

R104 ERJPA3F4701V — LOGIC_3V3
R105 ERJ-3EKF1001V — POLLEX
C103 C0603C103K5RACTU — LOGIC_GND

R106 ERJPA3F4701V — LOGIC_3V3
R107 ERJ-3EKF1001V — DIGITUS_MEDIUS
C104 C0603C103K5RACTU — LOGIC_GND

R108 ERJPA3F4701V — LOGIC_3V3
R109 ERJ-3EKF1001V — DIGITUS_ANULARIS
C105 C0603C103K5RACTU — LOGIC_GND

LOGIC_GND ×4
CN101 1990067

DIGITUS MINIMUS

0-30V OUT LINE RADIUS

Pull-down na GPIO
R112 ERJ2RKF4702X — LOGIC_GND
Obmedzenie prúdu LED
R110 ERJ3EKF3300V
OS_Trapenzium
U101 AQY211EH
LOGIC_GND
D101 SMBJ26CA_C83332

Pull-down na GPIO
R111 ERJ2RKF4702X — LOGIC_GND
Obmedzenie prúdu LED
R113 ERJ3EKF3300V
OS_Trapezoideum
U102 AQY211EH
LOGIC_GND
D102 SMBJ26CA_C83332

Pull-down na GPIO
R114 ERJ2RKF4702X — LOGIC_GND
Obmedzenie prúdu LED
R115 ERJ3EKF3300V
OS_Capitatum
U104 AQY211EH
LOGIC_GND
D103 SMBJ26CA_C83332

Pull-down na GPIO
R116 ERJ2RKF4702X — LOGIC_GND
Obmedzenie prúdu LED
R117 ERJ3EKF3300V
OS_Hamatum
U105 AQY211EH
LOGIC_GND
D104 SMBJ26CA_C83332

CN102 1990067

ESP32-S3-WROOM-1U-N16R8

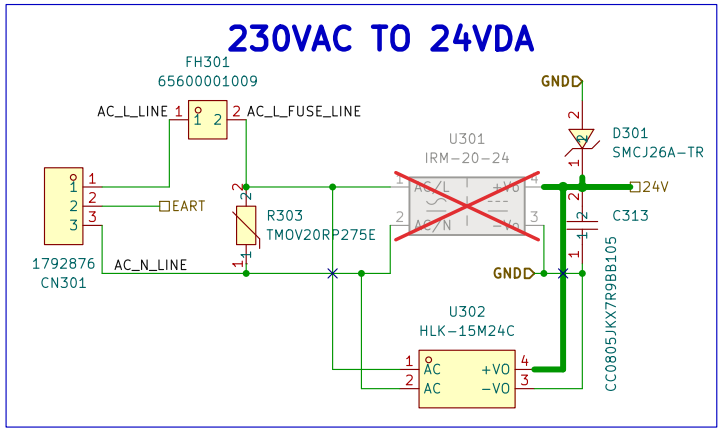
	ADC1	I2C_SCL	I2C_SCL
	ADC2	I2C_SDA	I2C_SDA
POLLEX	DI1_LOGIC	JTAG TDI	
INDEX	DI2_LOGIC		
DIGITUS_MEDIUS	DI3_LOGIC	JTAG TMS	
DIGITUS_ANULARIS	DI4_LOGIC	RGB_DATA	RGB_DATA
	DI_AUX1	SPI_CS1	SPI_SC1
OS_Trapenzium	DO1_CMD	SPI_CS2	SPI_CS2
OS_Trapezoideum	DO2_CMD	SPI_MISO	SPI_MISO
OS_Capitatum	DO3_CMD	SPI_MOSI	SPI_MOSI
OS_Hamatum	DO4_CMD	SPI_SCK	SPI_SCK
		SPI_DC	SPI_DC
	DO_FAULT_INT	UART0_RX	
	ESP_BOOT	UART0_TX	
RS485_A	RS485_A	UART_RX	UART_RX
RS485_B	RS485_B	UART_TX	UART_TX
LOGIC_GND	LOGIC_GND	CHASIS	CHASSIS_GND
LOGIC_3V3	LOGIC_3V3		

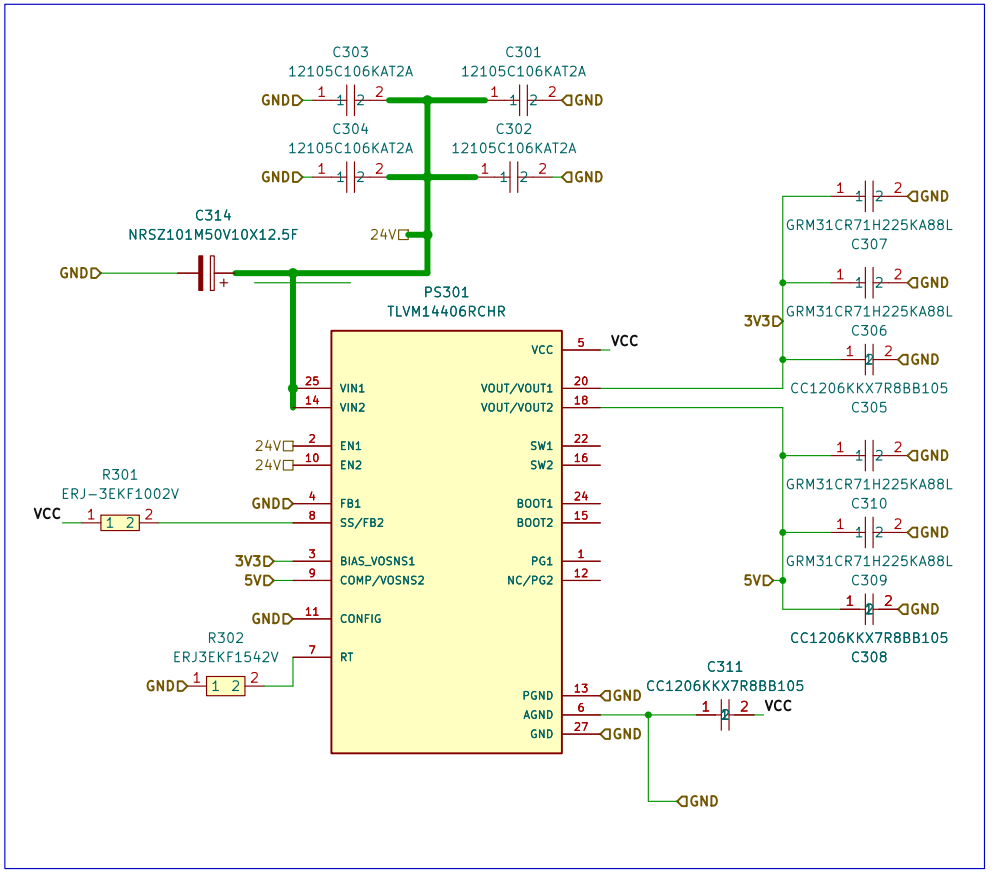
R121 MBB02070Z0000ZC100 — LOGIC_3V3 → 3V3
R119 — 5V_LOGIC → 5V
MBB02070Z0000ZC100 — 24V_LOGIC → 24V
CHASSIS_GND → EART
LOGIC_GND → GND
36V_TO_5V+3V3

RGB

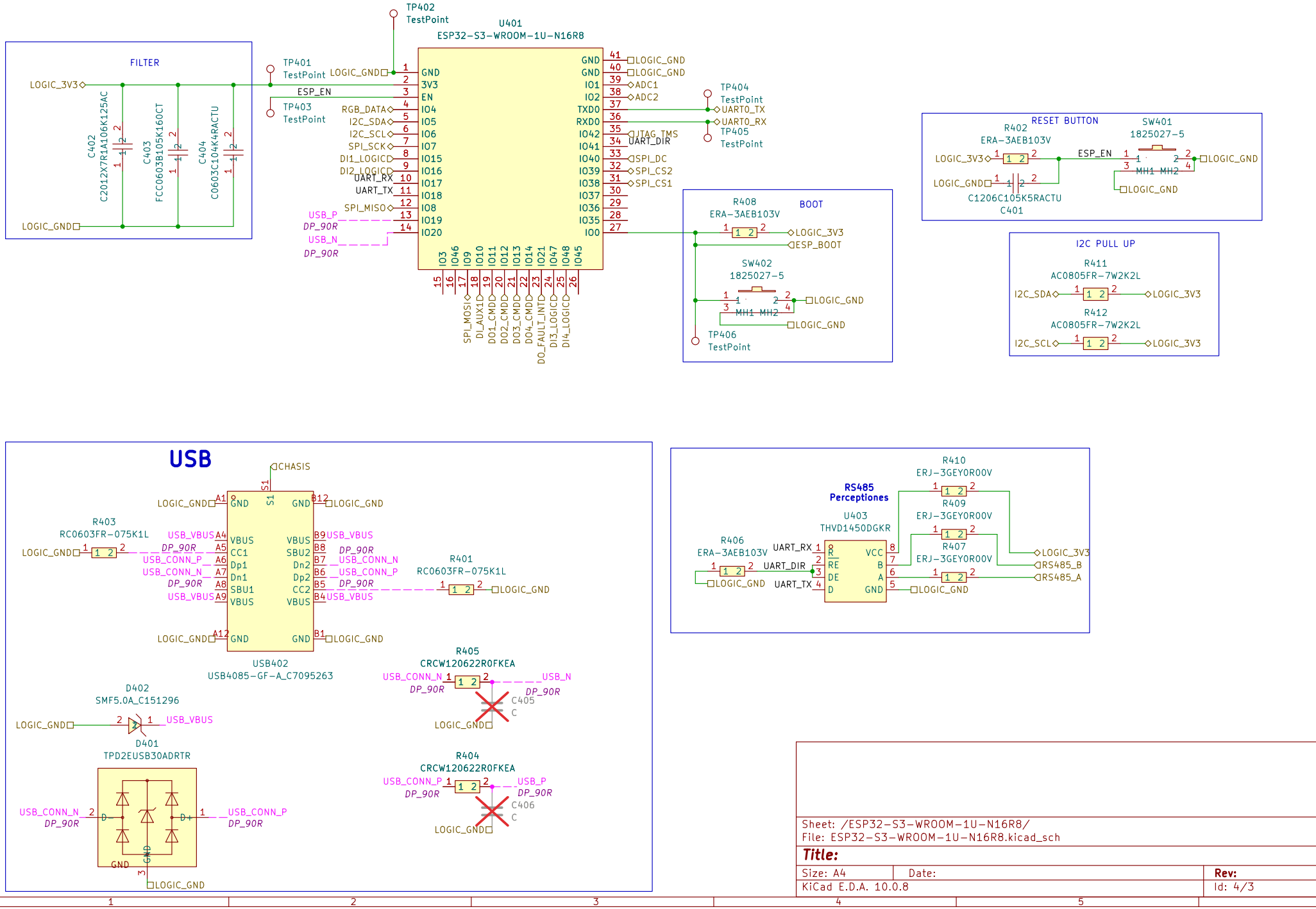
U103 SN74AHCT1G125DBVR
LOGIC_GND, OE, A, GND, VCC, Y
C101 C0805C104K5RACTU_C3018562 — 5V_LOGIC
RGB_DATA
R118 ERA-3AEB103V — LOGIC_GND
R101 ERJ3EKF3300V — LOGIC_GND
5V_LOGIC, LOGIC_GND
CN104 1989751

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Title:			
Size: A3	Date:		**Rev:**
KiCad E.D.A. 10.0.8			Id: 1/3





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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 3/3


FILTER
LOGIC_3V3
LOGIC_GND
C402
C2012X7R1A106K125AC
C403
FCC0603B105K160CT
C404
C0603C104K4RACTU
TP402
TestPoint
U401
ESP32-S3-WROOM-1U-N16R8
TP401
TestPoint
LOGIC_GND
ESP_EN
TP403
TestPoint
RGB_DATA
I2C_SDA
I2C_SCL
SPI_SCK
DI1_LOGICD
DI2_LOGICD
UART_RX
UART_TX
SPI_MISO
USB_P
DP_90R
USB_N
DP_90R
GND
3V3
EN
IO4
IO5
IO6
IO7
IO15
IO16
IO17
IO18
IO8
IO19
IO20
IO3
IO46
IO9
IO10
IO11
IO12
IO13
IO14
IO21
IO47
IO48
IO45
SPI_MOSI
DI_AUX1D
DO1_CMDD
DO2_CMDD
DO3_CMDD
DO4_CMDD
DO_FAULT_INTD
DI3_LOGICD
DI4_LOGICD
GND
GND
IO1
IO2
TXD0
RXD0
IO42
IO41
IO40
IO39
IO38
IO37
IO36
IO35
IO0
ADC1
ADC2
JTAG_TMS
UART_DIR
SPI_DC
SPI_CS2
SPI_CS1
TP404
TestPoint
UART0_TX
UART0_RX
TP405
TestPoint
R408
ERA-3AEB103V
BOOT
LOGIC_3V3
ESP_BOOT
SW402
1825027-5
LOGIC_GND
TP406
TestPoint
RESET BUTTON
R402
ERA-3AEB103V
SW401
1825027-5
LOGIC_3V3
ESP_EN
LOGIC_GND
MH1 MH2
C1206C105K5RACTU
C401
I2C PULL UP
R411
AC0805FR-7W2K2L
I2C_SDA
LOGIC_3V3
R412
AC0805FR-7W2K2L
I2C_SCL
LOGIC_3V3
USB
CHASIS
S1
LOGIC_GND
GND
VBUS
CC1
Dp1
Dn1
SBU1
SBU2
Dn2
Dp2
CC2
USB_VBUS
USB_CONN_P
USB_CONN_N
DP_90R
R403
RC0603FR-075K1L
R401
RC0603FR-075K1L
USB402
USB4085-GF-A_C7095263
D402
SMF5.0A_C151296
D401
TPD2EUSB30ADRTR
R405
CRCW120622R0FKEA
C405
R404
CRCW120622R0FKEA
C406
RS485
Perceptiones
U403
THVD1450DGKR
R406
ERA-3AEB103V
R410
ERJ-3GEY0R00V
R409
ERJ-3GEY0R00V
R407
ERJ-3GEY0R00V
UART_RX
UART_DIR
UART_TX
R
RE
DE
D
VCC
B
A
GND
RS485_B
RS485_A
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Date:
Rev:
KiCad E.D.A. 10.0.8
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